Natasha Mykytiuk (she/her)

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Open to Relocation

EDUCATION

University of British Columbia | Vancouver, BC **Expected May 2029**
Bachelor of Applied Science – Computer Engineering, Minor in Commerce
- Co-op Availability: 4, 8, 12, or 16 months

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, JavaScript, SystemVerilog, MATLAB
Embedded Systems & Hardware: STM32, FreeRTOS, CAN, FPGA, Altium, Arduino, Closed-Loop Control
Developer Tools & Platforms: Linux, Git, GitHub, Quartus Prime, Questa, Claude Code, Agent SDK
Software & Backend Development: React Native, Firebase, Firestore, Cloud Functions, LLM Integration, Figma
Engineering & Project Software: Aconex, Bluebeam, SOLIDWORKS, SharePoint, MS Office Suite, Photoshop

ENGINEERING DESIGN TEAMS

UBC Solar Racing | Vancouver, BC **Sep. 2026 – Present**
Embedded Firmware Developer
- Leading the **LCD display update** for the driver board (DRD), writing **C/C++ firmware** on **FreeRTOS** for an **STM32F103** that shows live telemetry data to the driver during races.
- Built **DRD health monitoring** that tracks fault flags and reports status over CAN to catch board failures.
- Decoded **CAN** messages from the telemetry system and updated the display in real time, splitting the work into separate FreeRTOS tasks for CAN input, health checks, and screen drawing.
- Collaborating with the electrical team to keep firmware in sync with **Altium** board revisions, debugging issues and validating changes on hardware.

WORK EXPERIENCE

Rangeland Engineering | Calgary, AB **May 2026 – Aug. 2026**
Engineering Intern
- Provided **project coordination** and **document control** across major industrial and infrastructure projects, including CNRL Horizon and the Trans Mountain Expansion Project.
- Performed quality checks on engineering drawings and technical documents using **Bluebeam** and **Aconex** prior to client transmittal.
- Tracked client comments and deliverable revisions, supporting timely turnaround and clear audit trails throughout review cycles.

TECHNICAL PROJECTS

Welicare (Dementia Care Mobile Application) | *Full-Stack Development, LLM Integration*
- Developing a **React Native** tablet application that helps caregivers provide personalized care for individuals living with dementia.
- Architecting a **Firebase** backend using Authentication, **Firestore**, and **Cloud Functions** with role-based access across shared care organizations.
- Developing an AI system for personalized activities, trivia, conversation prompts, and media recommendations with an accessibility-focused mobile interface.
- Using **Claude Code** to support development, debugging, and testing, with version control through **Git** and **GitHub**.
- Applying accessible design principles (ex. high contrast, large touch targets, legible typography), using software like **Photoshop** and **Figma**.

FPGA Keypad Scanner | *SystemVerilog, FPGA, Intel Quartus Prime*
- Designed a **SystemVerilog FSM** to scan a 4×4 matrix keypad on an **Intel MAX 10 FPGA**, outputting exactly one key code per press despite contact bounce.
- Integrated two-flop **synchronizers** and a **debounce** counter so held keys and switch bounce never register as extra presses.
- Built a **handshake interface** to send key codes to a second FPGA on a separate clock, holding data stable for the whole press so the receiver can sample it safely.
- Wrote an **SDC** clock constraint and **closed timing at 50 MHz** with 16 ns of setup slack, and verified the design with a self-checking testbench that models key bounce.